



Mahindra University Hyderabad
École Centrale School of Engineering,
Minor-1 Examination

Program: B.Tech Branch: Computation & Mathematics Year: III
Semester: I
Subject: Advanced Linear Algebra (MA3117)

Date: 12/09/2025
Time Duration: 1.5 Hours

Start Time: 02.00 PM
Max. Marks: 20

Instructions:

1. All questions are compulsory.
2. In Question 1, for a vector $x \in \mathbb{R}^{n \times 1}$, x^* is the transpose of x .

Q 1:

5 marks

Let $k \in \mathbb{R}$, $u, v \in \mathbb{R}^{n \times 1}$, and I denotes the Identity matrix of order n . If the matrix $(I + kuv^*)$ is the inverse of $(I + uv^*)$ then find the value of k .

Q 2:

5 marks

Find the matrix L in LU decomposition of the matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 1 \\ -6 & -1 & 2 \end{bmatrix}$.

Q 3 :

5 marks

Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ be a block matrix, where $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 6 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 3 \\ 4 & 8 \end{bmatrix}$, and $D = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Compute (M/A) , the Schur complement of A in M .

Q 4:

5 marks

Find the basis for the null space of the matrix A , where $A = \begin{bmatrix} 2 & -1 & 3 & 0 \\ 4 & 2 & 2 & 0 \\ 1 & -1 & 2 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}$.
